

DISRUPT-ED

A LOOK AT SOME OF THE MOST NOTABLE DISRUPTIONS THUS FAR

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The future usually lies in understanding the past, and we think it's no different when talking about tech disruption. Strictly speaking, disruption is really just a huge leap forward in a short amount of time, usually because of a new invention that makes something that was really difficult to do a heck-of-a-lot easier.

The term “disruptive technology” was first used by Clayton M. Christensen in his 1995 paper *Disruptive Technologies: Catching the Wave*, and was later made popular by his best-seller 1997 book *The Innovator's Dilemma*. According to him, a disruptive technology (or innovation) was one that would disrupt an existing market, and displace (if not ruin) an existing market leader. Also, since disruptive innovations are rarely ever profitable, and are usually high risk, it stands to reason why they usually come from outside the established industry players.

Now, strictly-speaking, everything from the discovery of fire to the invention of the wheel and then domesticating animals as load-pullers were disruptions in the evolution of the human race, but we're sticking to the modern age here, or basically the past 100 years or so, with emphasis on the last 30.

The first thing we need to understand is that disruptions are not always surprises that jump out of the shadows to strangle an existing market leader. Most often, they're known technologies

or inventions that have been deemed to be too niche, or too expensive, or just too weird and unwanted by the general population. The automobile is a prime example. The first steam-powered automobile was built in 1672 as a toy for the Emperor of China, but in the traditional understanding, many different road-worthy models of steam-powered vehicles started showing up between 1770 and 1800 in France and England. Even electric cars are not as recent as most would believe. Although many prototypes may have preceded it across the world, the Flocken Elektrowagen made by German inventor Andreas Flocken in 1888 is widely recognized as the first real electric car.

To say that horse and carriage manufacturers across the globe didn't have any inkling about the developments in automobiles would be a lie. In fact, automobiles were seen as toys for the wealthy (or the insane), and most people were scared of them because of the sheer amount of noise they made. One can easily imagine a meeting of the owners of horse and carriage companies, sipping their expensive drinks, sitting in their exclusive clubs making jokes about early automobile companies. However, history clearly tells us who had the last laugh...

In the same industry, today, electric cars are being opposed by everyone who makes a living from petrol and diesel vehicles. Petrol pump owners specifically don't want to accept the fact that electric cars are perhaps the disruption of their

industry. Performance and reliability of electric vehicles are increasing constantly, and when the tipping point comes, who's going to need a petrol pump? Petrol pump owners could embrace electric, and have quick car charging stations that would allow electric cars to also use petrol pumps. This might kill the need for innovations in electric cars for the charging and battery aspect, and allow pumps to survive. They won't, because there's no profit in it right now, which will force companies like Tesla to innovate with batteries and making it easy to charge from any A/C power outlet, and when the switch to electric starts to take off, the petrol pump owners are done for... and it will be too late for them to get with the times by then.

UP CLOSE AND PERSONAL

It was in the '80s that the idea of a personal computer took off. Before that, computers were huge mainframes that took up a room and did complex calculations that only a few people in the world could input. The IBM PC (amongst many others) made computers affordable for wealthier Americans. Then the price war started, and PCs kept becoming cheaper and cheaper, and soon almost every middle-class house across the developed world had one. Thus started the digital revolution. However, it wasn't until the Internet came along that the real digital disruption started...

What the PC did was take “computing” out of the hands of the “experts” and put it into the hands of everyone else. Now